

VASIL VASSILEV

www.vasilvassilev.cc • (734) 800-9772 • vvassilev515@gmail.com • github.com/vsicle

EDUCATION

University of Utah

May 2026

Bachelor of Science in Software Development • GPA: 3.8/4.0

Relevant Coursework: Database Systems, Computer Security, Web Development, Data Structures & Algorithms, Programming Languages

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, SQL, Rust, JavaScript, HTML/CSS

Frameworks & Libraries: FastAPI, React.js, Next.js, Django, PyQt6, .NET

Databases: MySQL, TimescaleDB

Tools & Platforms: Git, Docker, AWS (Lambda, EC2, S3), Linux/Unix, GitLab CI/CD, Grafana, n8n, Wireshark

AI & Automation: Retell AI API, Twilio API, Claude Code, Codex CLI, AMP code, Gemini API

Practices: OOP, REST APIs, Agile/Scrum, CI/CD, Test-Driven Development, Microservices, Database Design, Fuzz-testing

PROFESSIONAL EXPERIENCE

Software Engineering Intern / Capstone (Infrastructure & Networking) | L3Harris Technologies

2025 - 2026

- Architected **distributed network testing platform** using **Rust** and **Python** for military satellite communication systems, extending the open-source **TRex traffic generator** to coordinate unlimited chassis across geographically dispersed networks
- Engineered configurable **in-band/out-of-band data transfer system** with **GPS time synchronization** for sub-millisecond latency analysis, plus resilient failover mechanisms for continuous operation under unstable satellite links
- Built **real-time performance dashboards** integrating **Grafana** with **TimescaleDB** for time-series analytics, implemented **GitLab CI/CD** pipeline, and collaborated across cross-functional teams in **Agile** sprints with structured code reviews
- Diagnosed and resolved critical infrastructure deployment issues including misconfigured hardware causing **DDoS conditions**, providing technical consultation to IT teams on proper network testing equipment configuration

Founder | AI Voice Receptionist Agency (Self-Employed)

2025 - Present

- Built and deployed **AI-powered phone receptionist systems** for law firms and dental providers using **Retell AI API** and **Twilio**, handling 24/7 call answering, lead qualification, consultation booking, and attorney call routing
- Designed automation infrastructure with **n8n** connecting voice system webhooks to CRM platforms with automated follow-up communications and lead routing notifications

Founder & Lead Technician | Mobile Knife Sharpening (Self-Employed)

2020 - 2022

- Founded and grew service business to profitability across multi-city territory, building **React.js** booking platform that drove **30% increase in conversions** and managing end-to-end operations from customer acquisition to service delivery

PROJECTS & DEVELOPMENT EXPERIENCE

2D Physics Engine | C++, CMake, OpenAI Codex CLI

2026

- Built physics simulation engine in **C++** handling **collision detection**, rigid body dynamics, and gravity, using a **closed autonomous feedback loop** with the Codex CLI for rapid prototyping across 41 commits

Custom Text Editor | Python, PyQt6, MVC Architecture, TDD

2026

- Developed cross-platform text editor from scratch with custom **tokenizer-based syntax highlighting** engine supporting 6 languages, file tree navigation with auto-refresh, and memory-optimized undo/redo system
- Architected comprehensive test suite (**600+ passing tests** with pytest) enabling rapid development while maintaining code stability across 3-week build cycle

Open Source Vulnerability Research | C++, AFL++, AddressSanitizer, GDB

2024 - 2025

- Discovered **5 novel security vulnerabilities** including buffer overflows and memory corruption in Stormlib C++ library through systematic **fuzzing**, analyzing crash dumps via **GDB** and reverse engineering to map exploitable execution paths

Full-Stack Database Application | C#, .NET, MySQL, REST APIs

2024 - 2025

- Built **C#/.NET** application with normalized **MySQL** schema, **ACID-compliant** transactions, **RESTful** CRUD operations, and role-based access control for a multi-role learning management system

Multiplayer Networked Game | C#, TCP/IP, Multithreading

2023

- Built **TCP networking library** and game server from scratch without external dependencies, handling **concurrent client connections** with **multithreading**, async I/O, real-time state synchronization, and collision detection

LEADERSHIP & ACHIEVEMENTS

- 5x World Cup Rock Climbing Competitor** | Competed internationally at elite level across multiple disciplines
- Head Coach, University of Utah Climbing Team (2023)** | Led team to **#1 National Ranking**
- 3x Bulgarian National Champion** (1:8 Scale RC Racing)
- 3rd Place Michigan Invention Convention** | Arduino alarm clock designed for the deaf community
- U.S. Citizen** | **Bilingual:** Native fluency in English and Bulgarian